Solution :-

1. Displacement will be zero when initial & final point is same. Distance can not be zero if location is changed.

2. At any instant, the magnitude of velocity abd speed will be equal.

3. Magnitude of displacement can not exceed distance.

4. In projectile motion, acceleration is constant and trajectory is parabolic.

5. → If direction changes with constant speed then acceleration can change.

→ If an object is projected upward then at maximum height it's velocity is zero but acceleration due to gravity acts.

→ If velocity is constant then acceleration is zero.

6. $t=\sqrt{\frac{2H}{g}}$ ⇒ It doesn't depend on mass.

7. As body moves upward it's velocity decreases & at maximum height point it's velocity becomes zero but acceleration due to gravity act's downward.

9. A particle can not have two positions at same time.

13. Motion of the object with negative acceleration. Between O to t_1 it moves in positive x-direction

Between t_1 to t_2 it moves in opposite direction.

14. $S = ut + \frac{1}{2}at^2$

$0.21 = \frac{1}{2} \times 9.8\ t^2$

$t \cong 0.2\,\text{sec}$

15. $y = 0 - \frac{1}{2}gt^2 = -4.9\ t^2$

$y \propto -t^2$ parabola

16. ⊙ When we calculate velocity from displacement-time curve we take slope, similarly for velocity-time curve slope gives acceleration.

⊙ When we represent motion by a graph it may be x-t, v-t or a-t hence, Q may represent time.

⊙ For uniform motion velocity-time graph should be a straight line parallel to time axis.

⊙ For uniform motion velocity is constant hence slope will be positive for x-t curve so P may represent displacement.

⊙ For uniformly accelerated motion, slope will be positive & P will represent velocity.

17. For the ball dropped from building

$u_1 = 0$

$v_1 = u_1 - gt$

$v_1 = -gt$

for the ball thrown up, $u_2 = 40$ m/s

$v_2 = u_2 - gt$

$= 40 - gt$

Relative velocity of one ball w.r.t. to other

$= v_1 - v_2$

$= -gt - (40 - gt)$

$= -40$ m/s

When we apply equations of rectilinear motion we must carefully apply sign on physical quantities.

18. $v_A = \frac{\Delta x}{\Delta t} = \frac{20-10}{2} = 5$ m/s

$v_B = \frac{\Delta x}{\Delta t} = \frac{20-40}{2} = -10$ m/s

$v_{BA} = v_B - v_A = -10 - 5 = -15$ m/s

19. ⊙ As left is coming in downward direction, its position is above the ground.

⊙ Lift reaches 4th floor and is about to stop hence motion is retarding in nature, $v < 0$ so $a > 0$ acceleration act in upward direction.

⊙ Velocity is in downward direction, so $v < 0$.

20. There is no meaning of rest or motion without observer. For a person sitting in a moving car, the car is at rest but for an observer attached to ground, car is in motion.

21. $a = \frac{vdv}{dx}$

$\frac{dv}{dx} = \frac{a}{v} = \frac{\text{constant}}{v}$

because $v\uparrow$. So, rate of change of velocity with respect to distance will decreases continuously.

22. In 1-D, there are only two directions (backward and forward, upward and downward) in which an object can move or these two directions can easily be specified by (+) and (–) signs.

23. $x = v_0 t + \frac{1}{2}(v - v_0)t$

$x = v_0 t + \frac{1}{2} vt - \frac{v_0 t}{2}$

$x = \left(\frac{v + v_0}{2}\right)t = \bar{v}t$

24. An object is said to be in free fall, when its acceleration is only due to gravity. Its speed may be zero or non-zero.

25. $\vec{v}_{AB} = \vec{v}_A - \vec{v}_B$, because $\theta = 180^\circ$

$|\vec{v}_{AB}| = v_A + v_B$

26. Given, $|v_A| = |v_B|$

$v_{AB} = v_A - v_A = 0 = v_{BA}$ (same direction)

Case-II : $\vec{V}_A = -\vec{V}_B$ (opposite direction)

$\vec{V}_{AB} = -2\vec{A}_B$

$|\vec{V}_{BA}| = 2|\vec{V}_B| = 2|\vec{V}_A|$

27. The stopping distance depends on initial velocity (v_0), braking capacity and deacceleration.

28. $\vec{v} = \frac{v + v_0}{2}$ is valid for constant acceleration only.

29. $\vec{a} = \frac{\text{change in velocity}}{\text{time}}$

direction of $\vec{a}$ = direction of change in velocity

Equation of motion is valid for uniform acceleration. It may be in both x and y direction.